Probability Rules & Conditional Probability

Section 1: General Addition Rule & Mutually Exclusive Events

1 If $P(A) = 0.40$, $P(B) = 0.35$, and $P(A \cap B) = 0.15$, what is $P(A \cup B)$?

a. 0.60

b. 0.75

c. 0.90

d. 0.25

2 Two events E and F are mutually exclusive (disjoint). If $P(E) = 0.28$ and $P(F) = 0.45$, what is $P(E \cap F)$?

a. 0

b. 0.126

c. 0.73

d. 0.17

3 Can two events with non-zero probabilities be both mutually exclusive and independent?

a. No, if they are mutually exclusive, the occurrence of one prevents the other, meaning they are dependent

b. Yes, mutually exclusive and independent are synonyms in probability

c. Only if the sum of their probabilities equals 1

d. Yes, as long as their intersection is zero

Section 2: Conditional Probability Formula

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4 If $P(A) = 0.50$ and $P(A \cap B) = 0.20$, what is the conditional probability $P(B \mid A)$?

a. 0.40

b. 0.10

c. 0.70

d. 0.25

5 In a high school, 60% of students take math, and 18% take both math and art. Given that a randomly chosen student takes math, what is the probability they also take art?

a. 0.30

b. 0.18

c. 0.42

d. 0.108

6 Which statement mathematically proves that two events A and B are independent?

a. $P(A \mid B) = P(A)$

b. $P(A \cup B) = P(A) + P(B)$

c. $P(A \cap B) = 0$

d. $P(A) = P(B)$

Section 3: Multiplication Rule and Tree Diagrams

7 A bag contains 5 red marbles and 3 blue marbles. Two marbles are drawn at random without replacement. What is the probability that both are red?

a. $\frac{5}{14}$

b. $\frac{25}{64}$

c. $\frac{5}{8}$

d. $\frac{10}{56}$

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8 If two events C and D are independent with $P(C) = 0.60$ and $P(D) = 0.70$, what is $P(C \cap D)$?

a. 0.42

b. 1.30

c. 0.10

d. 0.65

9 A factory has Machine A producing 60% of items (2% defective) and Machine B producing 40% (5% defective). What is the total probability that a randomly chosen item is defective?

a. 0.032

b. 0.070

c. 0.025

d. 0.012

Section 4: Bayes' Theorem in Context

10 Continuing the factory problem: if an item is found to be defective, what is the probability it came from Machine A?

a. $\frac{0.012}{0.032} = 0.375$

b. $\frac{0.60}{0.032}$

c. 0.60

d. 0.020

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11 A diagnostic test has 99% sensitivity ($P(+|D) = 0.99$) and 95% specificity ($P(-|D^c) = 0.95$). If a rare disease affects 1% of the population ($P(D) = 0.01$), what is the probability that a person who tests positive actually has the disease?

a. ≈ 0.166 (about 16.6%)

b. ≈ 0.990 (99%)

c. ≈ 0.950 (95%)

d. ≈ 0.500 (50%)

12 What is $P(A^c \cup B^c)$ equivalent to by De Morgan's Laws?

a. $1 - P(A \cap B)$

b. $1 - P(A \cup B)$

c. $P(A) + P(B)$

d. $P(A \cap B)$

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